

PASSIONATE FAN PRESENTS

POWER WASH 2600

GAME MANUAL



AN ADAPTION OF
POWER WASH SIMULATOR
FOR THE ATARI 2600

ATARI 2600

LEGACY

In **1977**, Atari released their **second generation** console, the Atari Video Computer System (VCS), later known as the Atari 2600.

It's designers had two goals for the system:

- Be able to **imitate** existing games.
- Be **flexible** so new games can be made for the console.

With only **128 bytes** of **RAM** and **4KB ROM**, the Atari 2600 went on to become one of the most **influential** consoles of all time with many of its games having a lasting **legacy** even to this day.

HOMEBREW GAMES

Despite its age, new games are being made for the console by the **Homebrew** community, pushing the original **design goals to their limits** by creating amazing games with ingenious code solutions.

Famously, released in 2010 by Ed Fries, **Halo 2600** is an adaptation of Halo: Combat Evolved by Bungie, testing the original design goals in the context for the sixth generation consoles, becoming the **inspiration** for this project.



DESCRIPTION

MOTIVATIONS

- Explore the use of **old technology** to see how **viable** they are still **today**.
- **Critique modern** technology and **freedom** by developing a piece of software for a **limited** system.
- **Go further** than before by challenging the Atari 2600 for a **ninth-generation** console game.
- Learn all about **low-level development** and use **assembly** to develop software.
- Develop **skills** to transfer to new sectors that may benefit from such **knowledge**.
- **Celebrate** old technology and its **impact** on computing.

POWER WASH SIMULATOR

A **ninth-generation console game** for the **Xbox Series X and PS5** consoles built in the **Unity Engine** by FuturLab and SquareEnix.

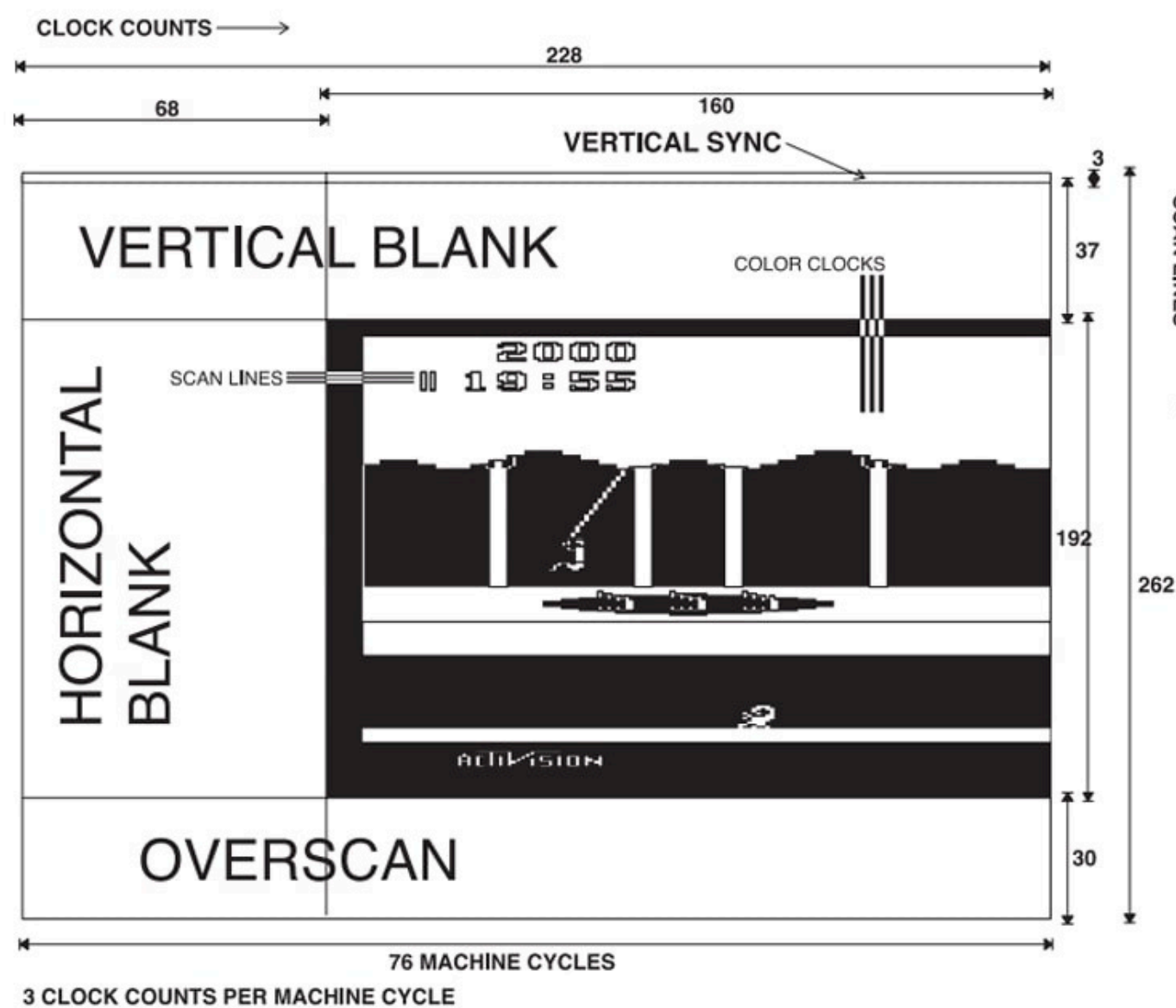
Players **satisfyingly blast** away **dirt** from **recognisable buildings and vehicles**, able to play with **friends** or solo.

This project adapts some recognisable aspects of the game, chosen for its **simplistic gameplay** and **art style that** translate well to the Atari 2600, being **achievable** within the **short development time** available.

RACING THE BEAM

CRT TVS

Unlike its competition, which used more conventional frame buffers, the Atari 2600 has a unique chip called the **Television Interface Adapter (TIA)** used to interface CRT TVs.



Developers need to “**Race the Beam**” to output anything on the console, counting **scanlines** and **machine cycles** with precision to position objects and avoid flickering issues.

SPRITES AND RESOLUTIONS

On top of that, there are only **5 sprites** that can be resized up to 8 times their size:

SPRITE	QUANTITY	RESOLUTION
Player	2	8 Bits
Missile	2	1 Bit
Ball	1	1 Bit

There are also **40 Playfield Graphics** per scanline stored in only **20 bits**, being the **lowest resolution** object, often used for walls and floors. Can be **mirrored** or **repeated**, **asymmetrical** fields need careful **timing**.

Collisions between these objects are detected by the **hardware**, making checking **trivial**!

6502 ASSEMBLY

```

sta PF2

lda MissileVel0
bmi .DrawRetic0

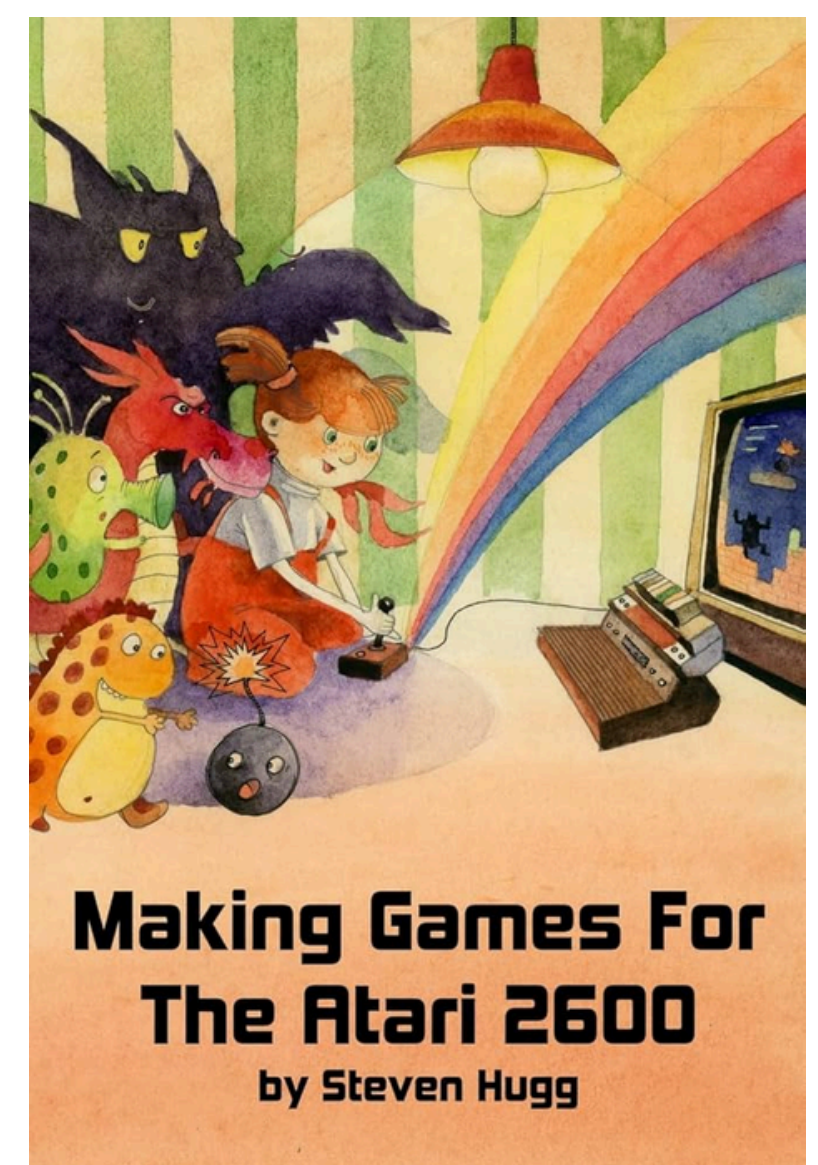
lda MissileY0
cmp Temp
bcc .EnableMissile0
jmp .DisableMissile0
.DrawRetic0
lda MissileY0
cmp Temp
beq .EnableMissile0
.DisableMissile0
lda #0
jmp .Finish0
.EnableMissile0
lda #2
.Finish0
sta ENAM0
    
```

Assembly for **MOS 6502 Processor**, a popular chip at the time, used in the Apple II, Commodore 64 and Nintendo Entertainment System (NES). It has **50 instructions** with variety of uses, each **costing** varying **machine cycles**. Other languages such as BatariBasic are available, but are not as **transferable**. **Hard to learn**, but **rewarding** to optimise and have fine control over the hardware.

8BITWORKSHOP

An **IDE** for Atari 2600 game development, complete with useful **macros**, **debugging** tools, **time analysis**, **emulator** (“Javatari”) and **learning** materials.

Developed by Steven Hugg, the accompanying book was an instrumental resource to **learning assembly** and **challenges** involved for Atari 2600 development.

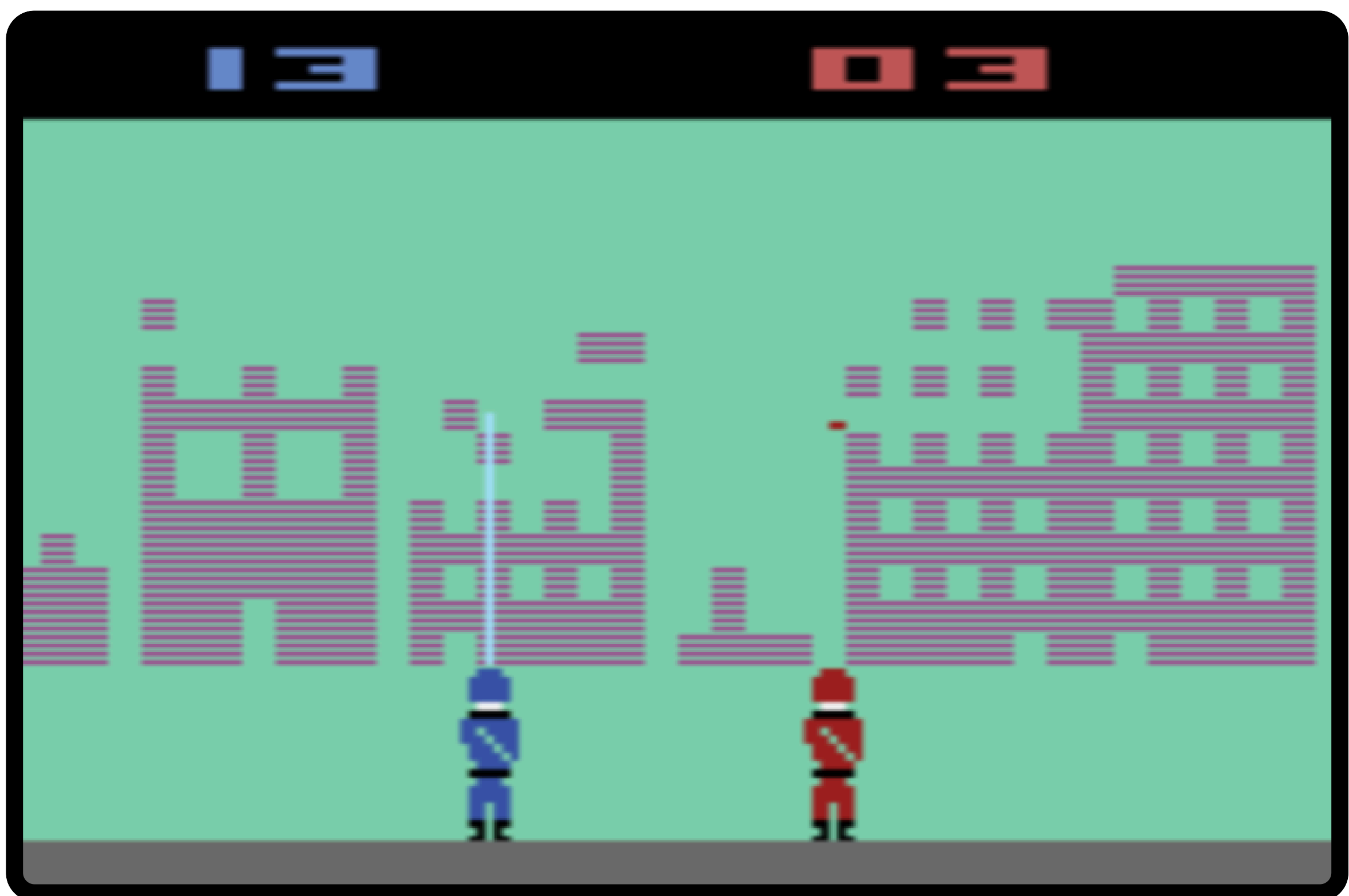


HOMEBREW TOOLS

Tools made by the **Homebrew community**, such as ‘**PlayerPal2600**’ to create **sprites and animations**, as well as ‘**Atari Sound Forger**’ to explore sounds, help to bring Atari 2600 game development to the **modern** day, making many aspects of development easier.

FEATURES

- 1. MOVEMENT AND AIMING** Take control of your character using the Joystick: pulling left to go left; pulling right to go right. Take aim by pulling the joystick up or down to move your reticle in the corresponding direction. Press the button to fire the washer! Release to stop. If you have used a thumbstick, you will be right at home!
- 2. CLEANING** Clean dirt away from the screen. Take aim at a piece of dirt, fire and see it satisfyingly wash away. Complete a section to increase your score at the top of the screen and hear a satisfying ding, just like the original! Be careful, sections 2-3-5-6 take more effort!
- 3. PROCEDURALLY GENERATED JOBS** Take aim and clean a variety of randomly appearing buildings and vehicles. Once you complete a round, new objects will appear generated using the game code itself! Continue the satisfaction of cleaning forever, or play other modes.

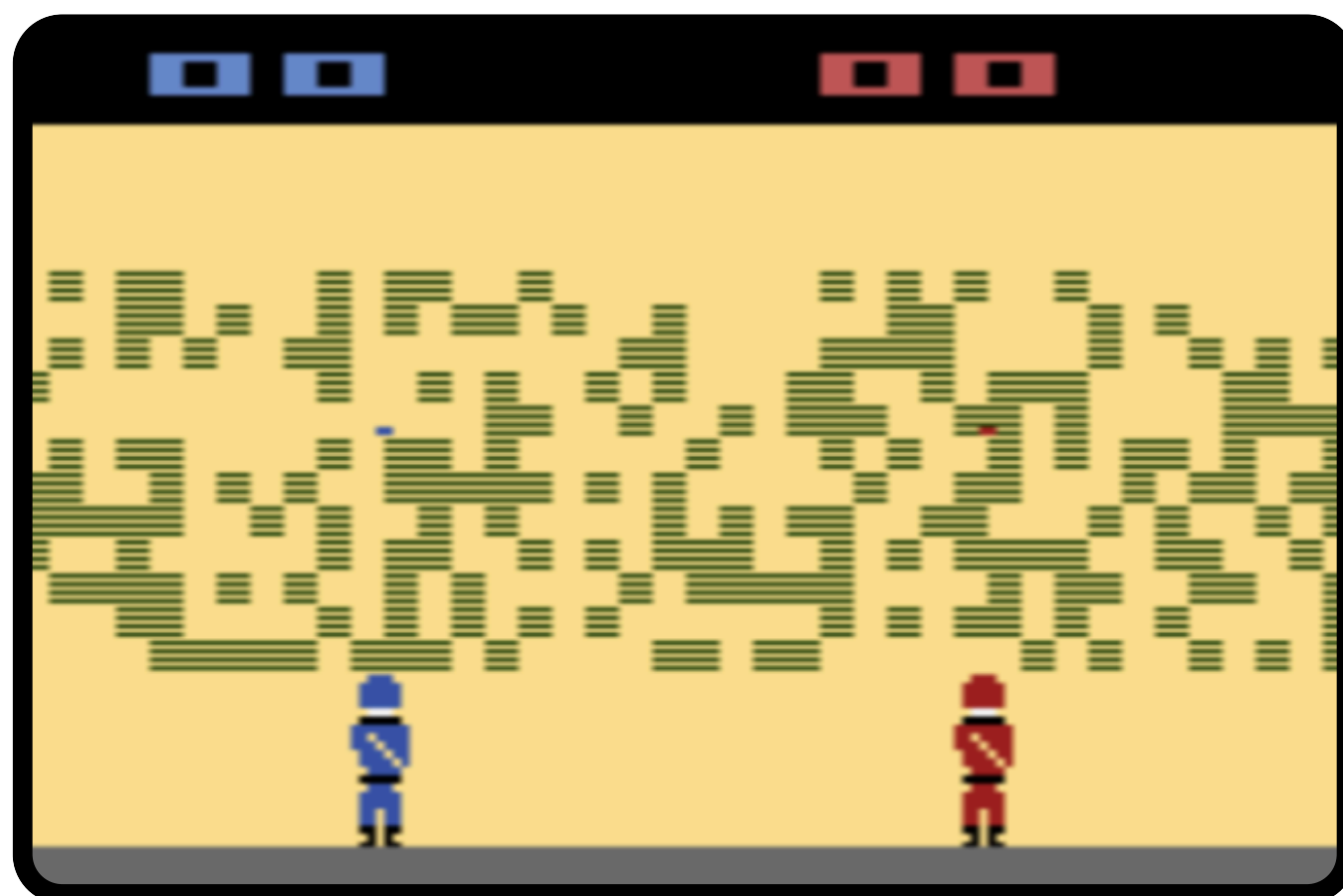


FEATURES

4. **CANVAS MODE** Like to draw? Flip the Color-BW switch to BW, and hit game-reset (or wait until the next round) to enter canvas mode! Here, the whole screen can be washed, letting you draw fun pictures. What can you draw?



5. **RANDOM MODE** Want a bit more challenge and be left guessing what comes next? Flip the switch to BW and flip P2 to expert mode to activate completely random generation! Here, you can see the game code itself become the dirt!



6. **COMPETE** Want to know who's the better Power Washer? Compete against a friend to see who can get the most score. Watch out, your friend may try to steal your points by completing a section before you do! Otherwise, you can work together and both be Power Washer champions

CONCLUSIONS

VIABLE

This project proves that the **Atari 2600** is still **capable** of achieving its original aims, going further by imitating a **ninth-generation** console game, Power Wash Simulator, on the **second-generation** console with this game, **Power Wash Simulator 2600**. Future work can go onto push the console further, or explore other old technology and discover new innovations from the past.

SKILLS

For any developer, learning **low-level languages** and having **constraint** is an **enormously beneficial** exercise, appreciating hardware and history more. Skills developed can be **transferred** to multiple sectors, such as **AI, graphics, etc**, to make minute **optimisations** for **significant performance** gains. This project hopes to inspire new developers to take on this challenge, learning from the **past** to innovate in the **future**.

TIPS FROM THE DEVELOPER



MATTHEW PRICKETT, THIRD YEAR SOFTWARE ENGINEERING STUDENT

"Making a game for the Atari 2600 has been one of the most fun experiences I have ever had as programmer, combining my love for games, challenge and coding. I would recommend it to anyone (willing enough to subject themselves to assembly). I cannot wait to return one day to improve this game and make new ones for new systems (NES I'm looking at you)."

"I hope you enjoy my game and thank you so much for playing!"